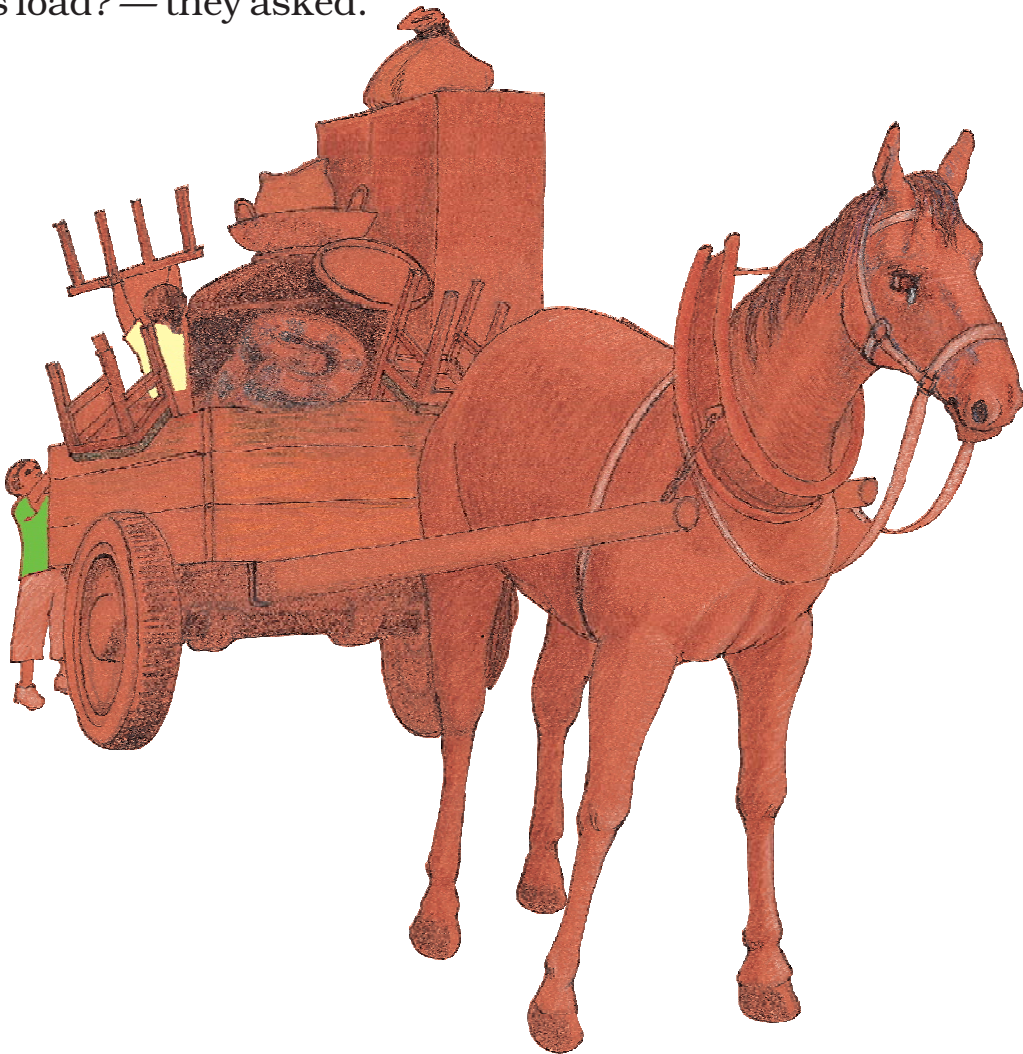
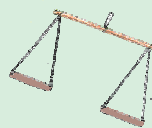
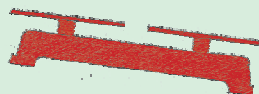
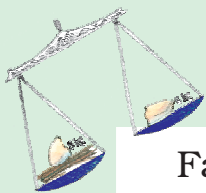


12 How Heavy? How Light?

Jaiju and Mannu were shifting house. They loaded all their things on a horse-cart. There were many things like — a water tank, five sacks of wheat, three tables, an almirah, four chairs, two mattresses, three sacks of rice, a bamboo ladder, pots and pans.

When they were ready to move, the horse refused to start. They wondered why. Their father said that this horse was not well and would not pull a load heavier than 700 kg. Oh! But how heavy is this load? — they asked.





Father gave them some idea of the weight of each thing.

- * Find out the total weight they had loaded on the cart.

Now they decided to remove a few things from the cart.

- * Which things should be removed so that the weight of the load is not more than 700 kgs?

Thing loaded	Weight
A sack of wheat	100 kg
A sack of rice	35 kg
Water tank	50 kg
Almirah	70 kg
A table	10 kg
A chair	5 kg
A mattress	20 kg
Bamboo ladder	10 kg
Pots and pans	10 kg

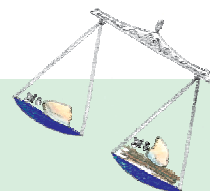
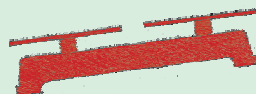
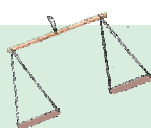


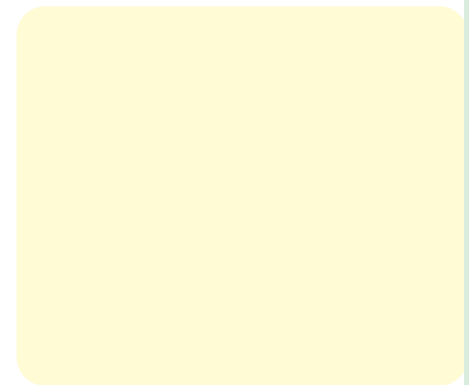
The things which were loaded on the cart were big in size and also very heavy. To measure the weight of such heavy and big things, we need a big balance.

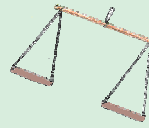
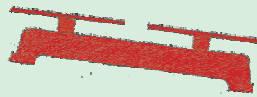
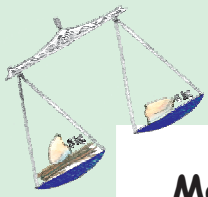
But Jaiju and Mannu wanted to make their own balance. They collected a few things — a stick, two lids and a thick thread. They made this balance.



- * Now you also make your own balance. Write down how you made it. Also draw a picture of your balance in the box below.







Making Weights

Do this activity in pairs. You need a balance, weights, a cake of soap, plastic packets, sand and rubber bands. You can also take help of an older person.

Get a new cake of soap. The packet will have the weight written on it. You can use this soap to make your own different weights.

The soap weighs _____ grams (g).

Take a small plastic packet.

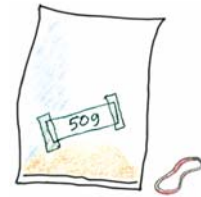
Put it in one pan of the balance. Put the soap in the other pan.

Slowly add sand to the packet till the pans are balanced.



Close the packet with a rubber band or string. Now stick a strip of paper and write '____ g' on it.

If you put the soap and the weight you just made together in a pan, how many grams will both these weigh? _____



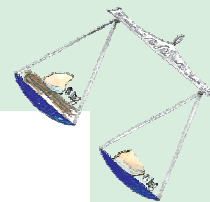
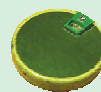
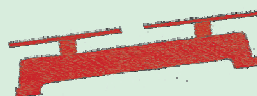
Now make different weights — 150 g, 200 g and 250 g. You can use soaps of different weights for this.



Also make some bigger weights of 500 g, 1000 g, and 750 g.



* Use your weights to weigh different things and write in your notebook.



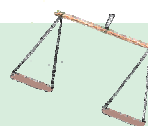
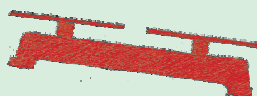
Practice Time

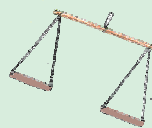
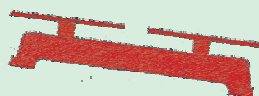
* Which pan of the balance will go down? Show by drawing an arrow.



* Is the weight on any of the pans equal to 1 kilogram? Mark it.

* How many grams are there in 1 kg?





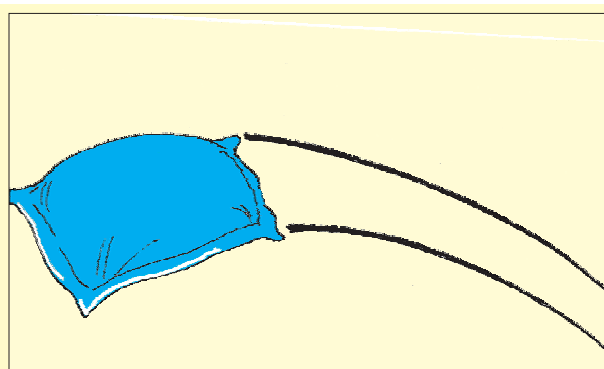
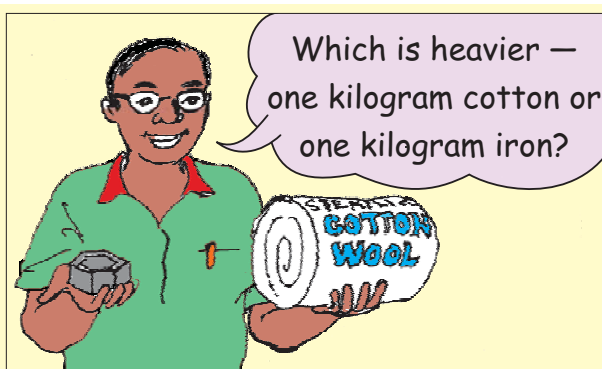
Grams and Kilograms

Name 5 things that we usually buy —

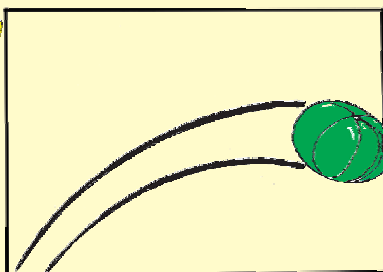


In grams	In kilograms
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.

Which is Heavier?

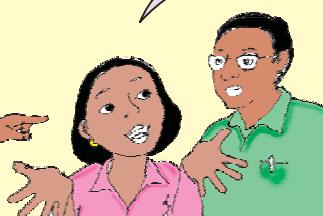
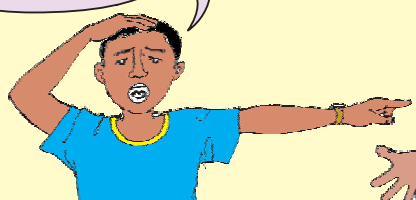


OUCH!

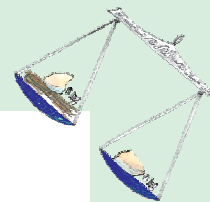
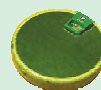
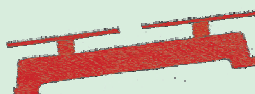


AAA... AAH!

Sir, she threw a heavy watermelon at me!



He threw a one kg pillow of cotton. So, I threw a one kg watermelon! Yesterday you said that the weight of 1 kg cotton and 1 kg melon is equal.



Dinesan Went Shopping

Dinesan went to a shop and bought some things.

His younger brother cut the end of the bill where the weights were written.

* Guess and write the weight of each thing he bought in g or kg.



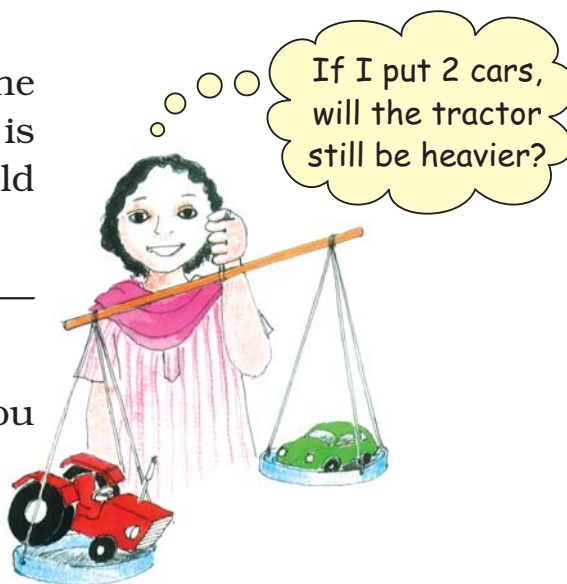
Items	Weight	
Rice	5	
Sugar	1	
Mustard seeds	10	
Wheat	3	
Dal	500	
Tea	250	
Pepper	25	

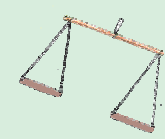
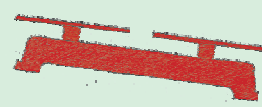
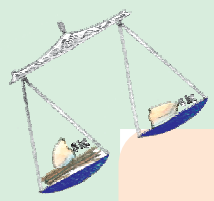
Car and Tractor

Ritu is weighing her toys. She wants to know if her tractor is heavier than her car. How would you help her to find out quickly?

Guess which is the heaviest — a real car, a bus or a tractor?

Which is the heaviest thing you have seen?





Elephant's Weight

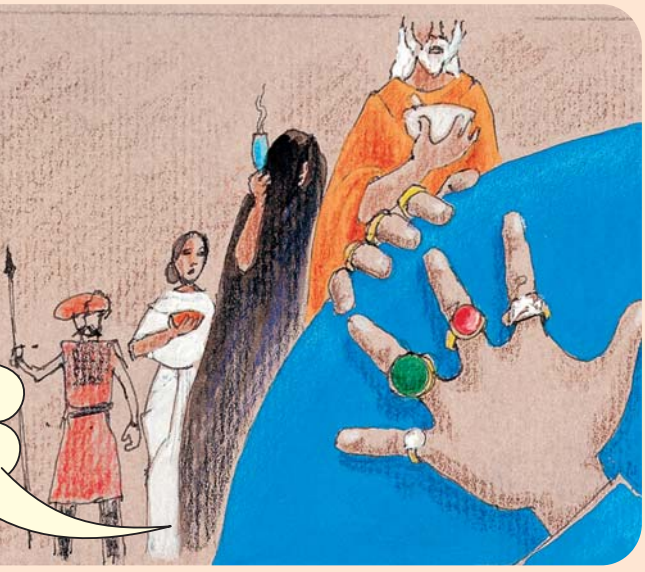
Once a king had pain in his stomach. None of the palace doctors could cure his pain.

Oh, no! I cannot bear this pain.



The king then said:

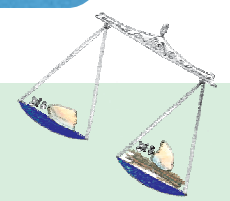
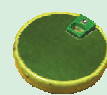
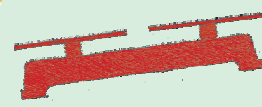
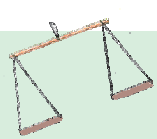
Anyone who cures my pain will get gold equal to the weight of an elephant.



On hearing this, doctors from all over the country came. But only Dr. Vaidika could cure him.

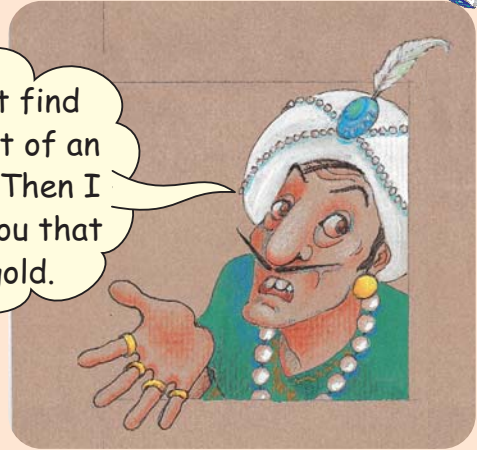
Oh great! My pain has gone. Thank you, Dr. Vaidika.

So, can I have my reward now, sir?



But, the greedy king didn't want to give her the gold. So, he thought of a trick.

OK, first find the weight of an elephant. Then I will give you that much gold.



Vaidika was unhappy when she reached home. She told her daughter the whole story.

How can I weigh an elephant?
Where will I get such a big balance?

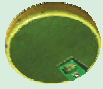
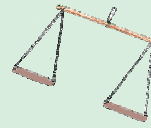
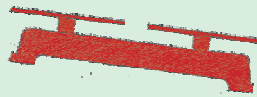
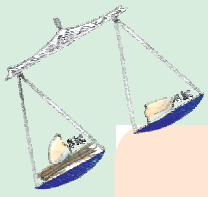
Don't worry Ma. I have an idea tell the king to arrange an elephant and a big boat.



Next morning, Dr. Vaidika invited the king near a river. The king came with an elephant and a big boat.

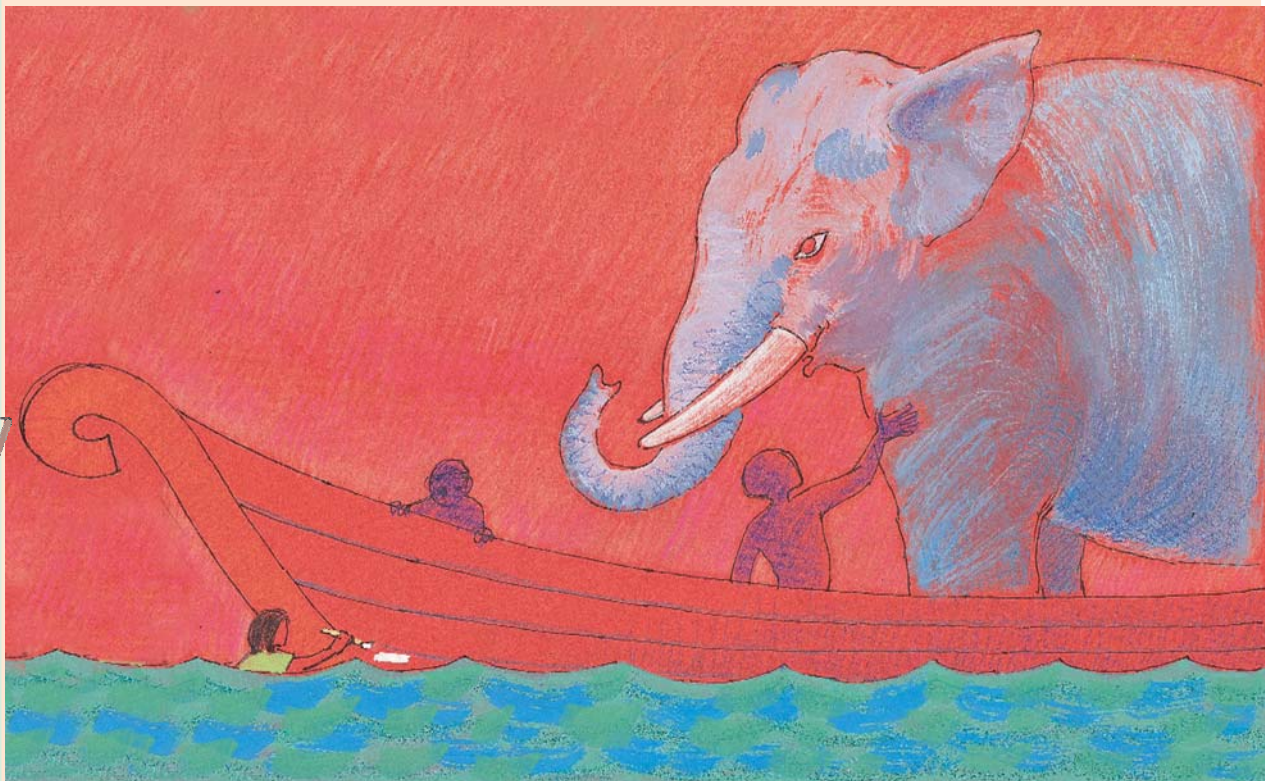
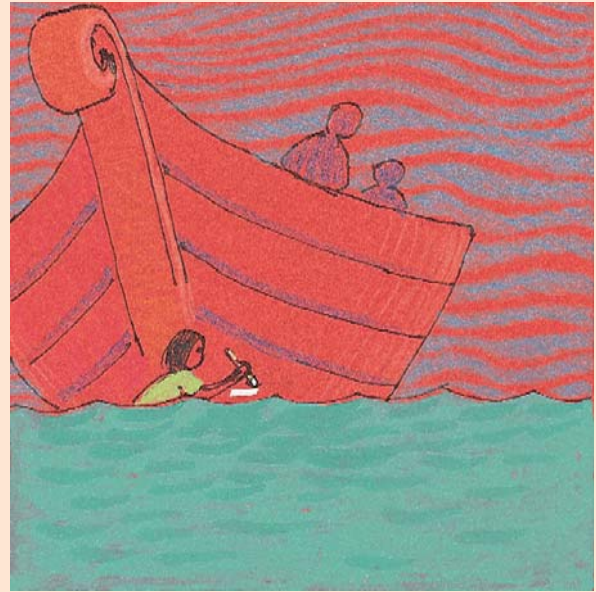
I think she is a fool.
How will she weigh an elephant with a boat!



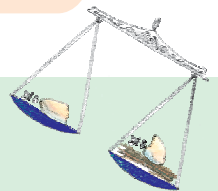
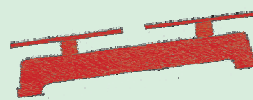
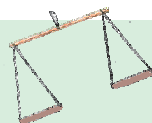


Vaidika's daughter went into the river. She marked on the boat how much it sank in the river.

Then she asked them to bring the elephant into the boat. The boat sank deeper. So she marked the new water level on the boat.



Now imagine what happened next and complete the story. Discuss with your friends how Vaidika's daughter found the weight of the elephant.



How Much the Chair Weighs

Anamika wants to weigh this chair using the weighing machine.



Can you suggest a way for doing this?



Broken Stones

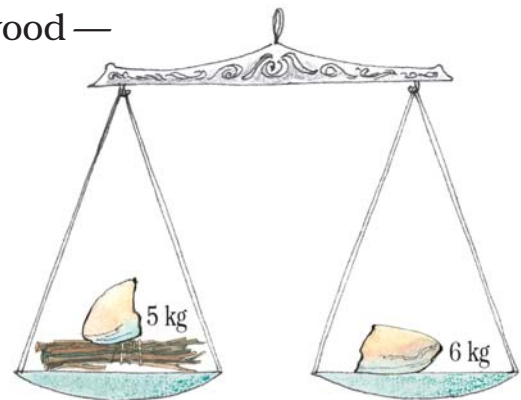
Abdu sells firewood. There was a stone in his shop which weighed 13 kg. He used it to weigh firewood.

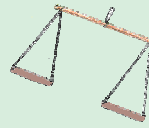
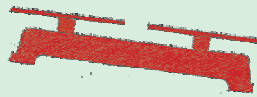
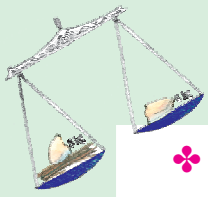
One day the stone fell down and broke into three pieces which weighed – 2 kg, 5 kg and 6 kg.



But Abdu was very smart. He used those three pieces to weigh firewood of all weights — from 1 kg to 13 kg.

Here is how Abdu weighed 1 kg of firewood —





✿ Now you show how Abdu will use these stone pieces to weigh —



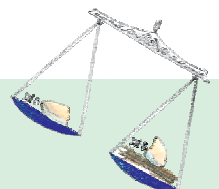
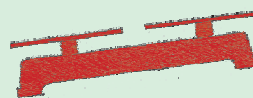
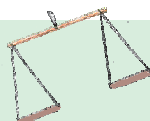
a) 4 kg of firewood



b) 3 kg of firewood



c) 7 kg of firewood



Post Office

Have you ever been to a post office? _____

What different things do people go there for? _____

How much does a postcard cost? _____

How much does an inland letter cost? _____

Hkkjrh; Mkd  INDIA POST	Postal Rates 	
--	---	---

Postal Items	Postal Rates (in Rs)
Single post card	0.50
Printed post card	6.00
Inland Letter	2.50
Letter weighing	
i) 20 grams or less	5.00
ii) For every additional 20 grams	2.00
Parcel weighing	
i) 50 grams or less	5.00
ii) For every additional 50 grams	3.00



Look at the postal rates given in the chart.

1. How much will you have to pay for stamps on a letter weighing 50 grams? _____
2. Akash wants to send a parcel of the Math Magic textbook to his friend Rani in Chennai. The book weighs 200 g. See the chart to find the cost of posting the book. _____



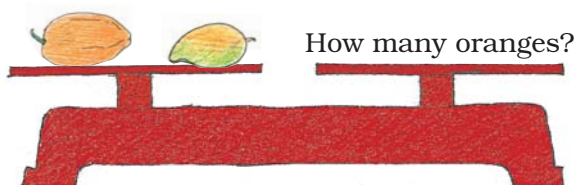
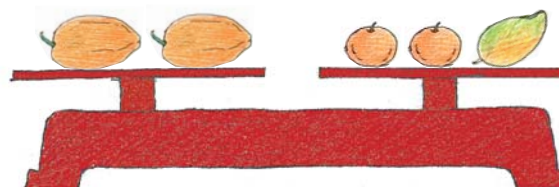
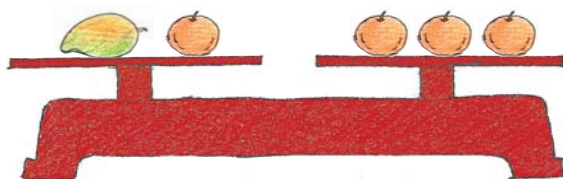


Puzzle

How Many Oranges?

All oranges have equal weight. The two papayas have the same weight. The weights in the first and second balances are equal.

How many oranges balance the weight in the third?



How many oranges?

Find That Marble

There are 3 marbles of the same size but one marble is slightly heavier or lighter than the other two. Can you find which is that marble and if it is heavier or lighter? You can use a balance only two times.

